

DIMITRIOS PANAGIOTIS EMMANOUILIDIS

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RESEARCH INTERESTS

MLOps, LLM System Orchestration, Local Inference Optimization, Multi-Agent Systems, Autonomous Vehicle Simulation, Retrieval-Augmented Generation (RAG).

EDUCATION

B.Sc. in Software Development

Expected July 2026

New York College in partnership with University of Greater Manchester — Thessaloniki, Greece

- **Current Standing:** Level 6 Weighted Average: 62.83% (Upper Second-Class Honors Equivalent) | Cumulative Average: 53.39%
- **Undergraduate Thesis:** “ML/LLM System Orchestration” (Defended May 29, 2026)
 - Architected a multi-agent AI system designed for an autonomous vehicle simulator.
 - Implemented real-time sensor fusion to process and analyze simulated environmental data.
 - Evaluated model latency and decision-making accuracy within a simulated real-time constrained environment.
- **Relevant Coursework:**
 - *Research & Analysis:* Undergraduate Research Project (76%), Contemporary Problem Analysis.
 - *Software Engineering:* Introduction to Software Development (76%), Enterprise Systems Development (67%), Data Structures and Algorithms, Object Oriented Programming, Open Source Software Development.
 - *Systems & Data:* Introduction to Networking (73%), Database Design, Secure Application Development, Server Application Configuration and Management.

RESEARCH & TECHNICAL PROJECTS

Offline Local Inference & Retrieval Application (Startup Concept)

In Development

Lead Developer

- Developing a frictionless, unified application for local LLM inference and off-grid knowledge retrieval.
- Integrating a heavily quantized local LLM optimized for execution on consumer hardware, spanning mobile devices and eGPU-accelerated local setups.
- Architecting a built-in Retrieval-Augmented Generation (RAG) pipeline connecting the model directly to massive offline databases (e.g., Wikipedia, medical guides).
- This architecture is designed to allow the LLM to query offline archives and generate factual responses without relying on external APIs or manual terminal configurations.

Local AI Infrastructure & Orchestration Deployment

Late 2025 – Present

Independent Research

- Engineered a high-performance local AI execution environment utilizing a Beelink GTi13 Ultra paired with an external NVIDIA RTX 5070 GPU.
- Deployed and optimized local orchestration frameworks, including LangGraph and Ollama, focusing on VRAM allocation and local inference optimization.

Curated Map-Based Travel Application

June 2026 – Present

Lead Developer

- Developing a mobile application targeting city break travel planning and memory curation.
- Integrating map-based memory tracking features, including a “storybook” recap system and a “Travel Wrapped” data visualization tool.
- Designing the backend architecture to support real-time data fetching and user authentication.

TECHNICAL SKILLS

- **Programming Languages:** Python.
- **AI & Machine Learning:** LangGraph, Multi-Agent Orchestration, Local LLM Inference, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), Sensor Fusion.
- **Infrastructure & Deployment:** Docker, VRAM Management, Hardware Optimization (eGPU configurations), MLOps.

CERTIFICATIONS

- AWS Certified Generative AI Developer – Professional (AIP-C01) | Expected January 2027
- AWS Certified Machine Learning Engineer – Associate (MLA-C01) | Expected October 2026
- MLOps Zoomcamp (DataTalks.Club) | Expected September 2026
- DeepLearning.AI Machine Learning Specialization | Expected August 2026

LANGUAGES

- **Greek:** Native | **English:** C2 Level Certification